Chapter 12 — Nitrogen and Sulfur

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Nitrogen is unreactive due to: (a) Weak bond (b) Strong triple bond (c) Polarity (d) High solubility

2. NH_3 is a base because it: (a) Accepts H^+ (b) Releases OH^- directly (c) Donates H^+ (d) Accepts e^-

3. NH_4^+ shape: (a) Linear (b) Planar (c) Pyramidal (d) Tetrahedral

4. Nitrification converts NH_4^+ to: (a) N_2 (b) NH_3 (c) NO_3- (d) NO_2-

5. Denitrification converts nitrate to: (a) O_2 (b) NH_3 (c) N_2 (d) NO_3-

6. Sulfur occurs as: (a) S_2 chains (b) S_4 cubes (c) S_8 rings (d) S atoms only

7. Sulfur in H_2S has oxidation state: (a) -2 (b) 0 (c) +4 (d) +6

8. Sulfur is used in: (a) Photography only (b) Vulcanization (c) Refrigeration (d) Water softening

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is nitrogen used as a protective blanket?

2. How does ammonia act as a Bronsted-Lowry base?

3. Why is the ammonium ion tetrahedral?

4. How does nitrification increase soil fertility?

5. Why does sulfur show variable oxidation states?

6. How does sulfur behave as an oxidizing agent with metals?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the inertness of nitrogen and the basicity and preparation of ammonia. **(4)**

(b) Explain nitrification and denitrification. **(4)**

Q.4 (a) Explain the reactivity and oxidation states of sulfur. **(4)**

(b) Describe reactions of sulfur with metals, oxygen and hydrogen, and its uses. **(4)**

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